

AI MENTAL HEALTH ASSISTANCE

ANA – 2

Report On Demographic Breakdown Analysis

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Dataset used: Adult Psychiatric Morbidity survey (apms) 2014

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Submitted By: zeno talent

Demographic Breakdown Analysis

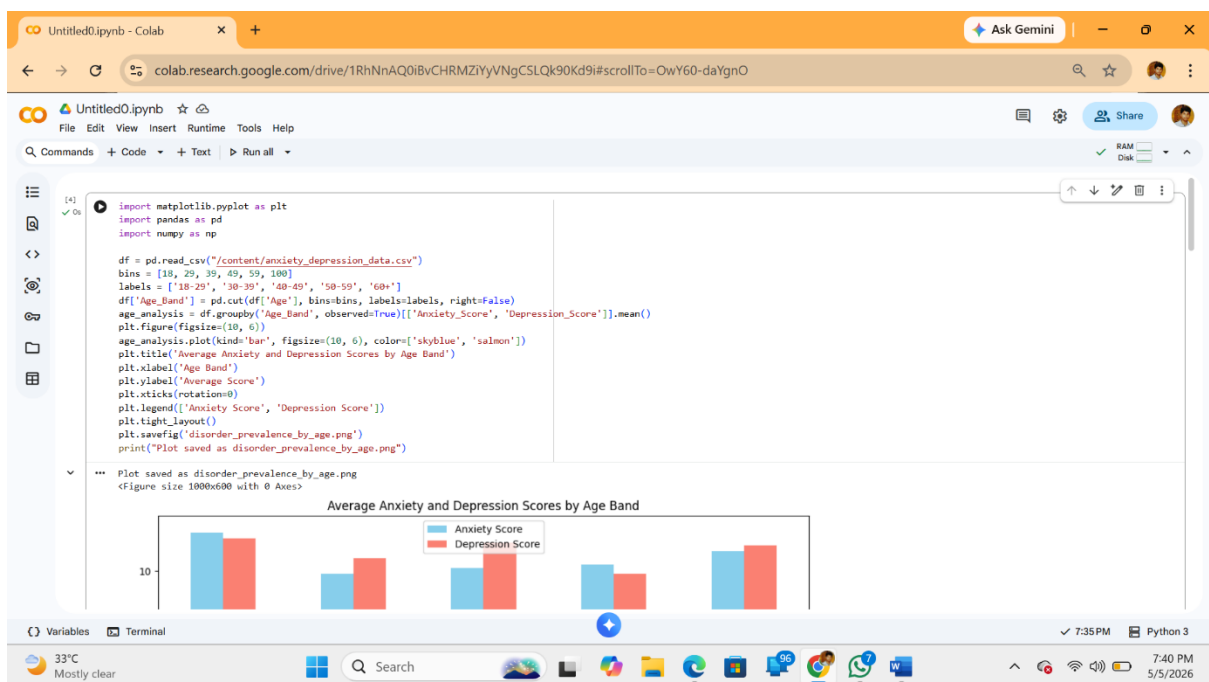
The analysis identifies how mental health metrics fluctuate across different population segments. Understanding these trends can help in pinpointing which groups might be more susceptible to elevated stress or mental health burdens.

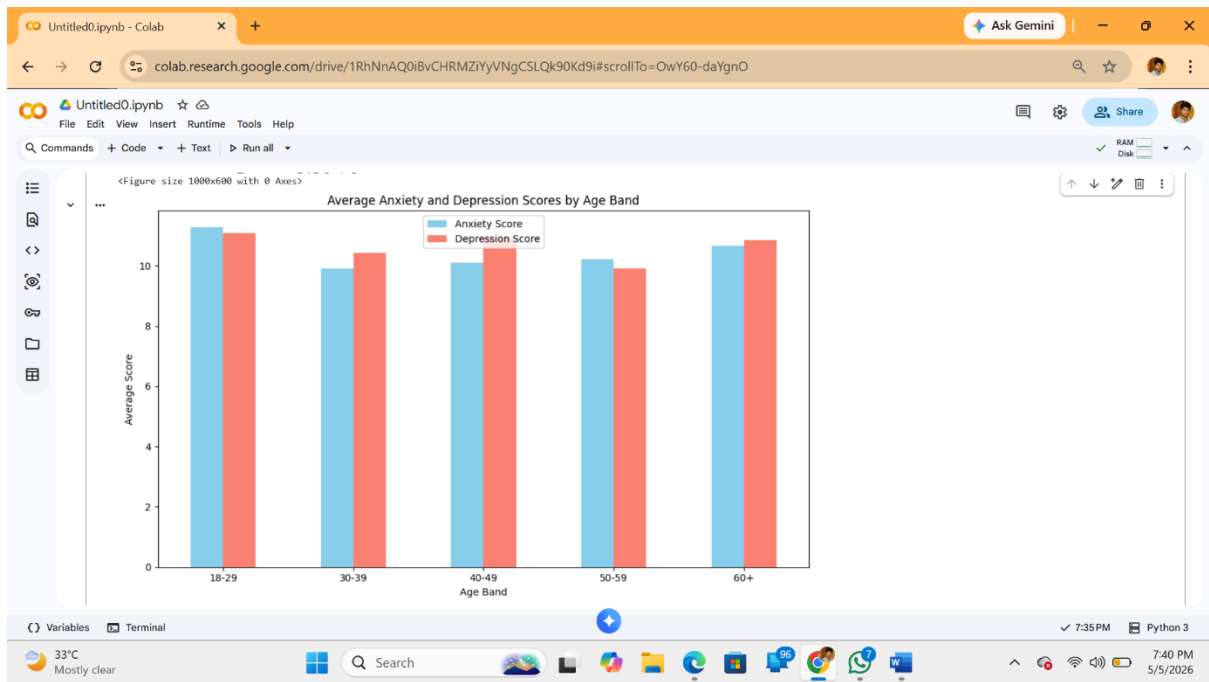
Technologies Used

- **Pandas:** Used for aggregating data and calculating the mean Anxiety_Score and Depression_Score across categories.
- **Matplotlib:** Used to generate the grouped bar charts, which facilitate side-by-side comparison of the two mental health metrics.

Analysis of Disorder Prevalence by Age Band

This analysis examines the prevalence of mental health disorders—specifically anxiety and depression—across different age demographics. By segmenting the dataset into age bands, we can identify trends and potential patterns in mental health scores, helping to highlight whether certain life stages are associated with higher psychological burdens.

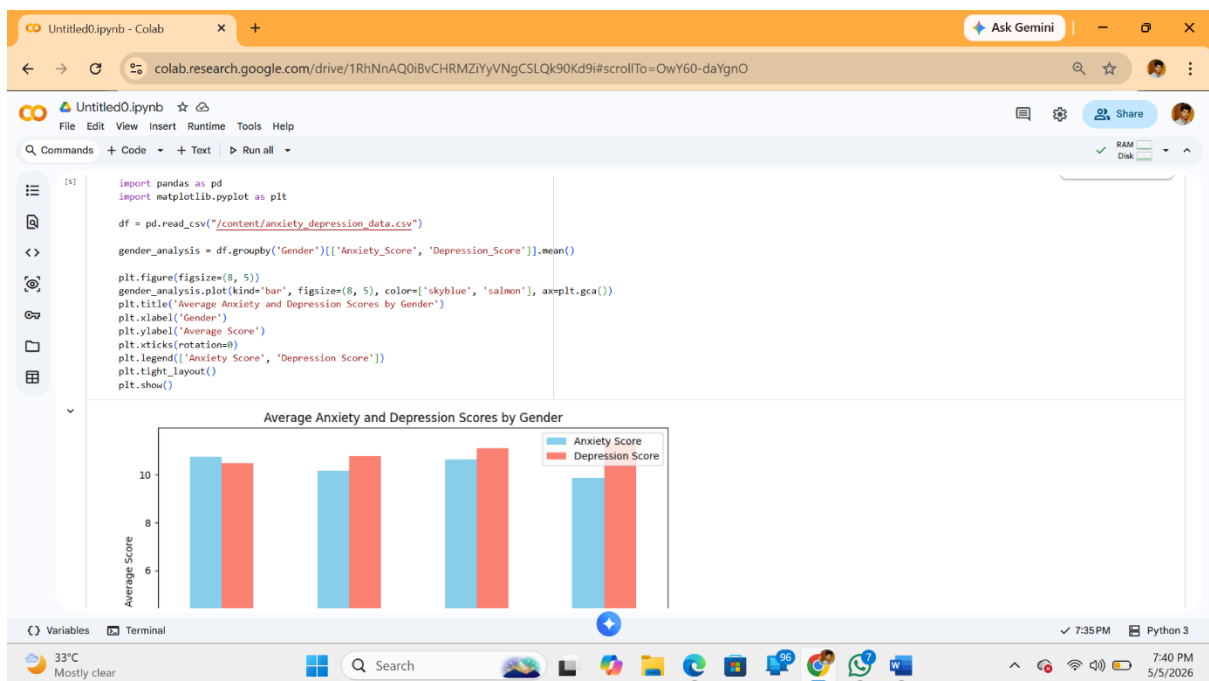


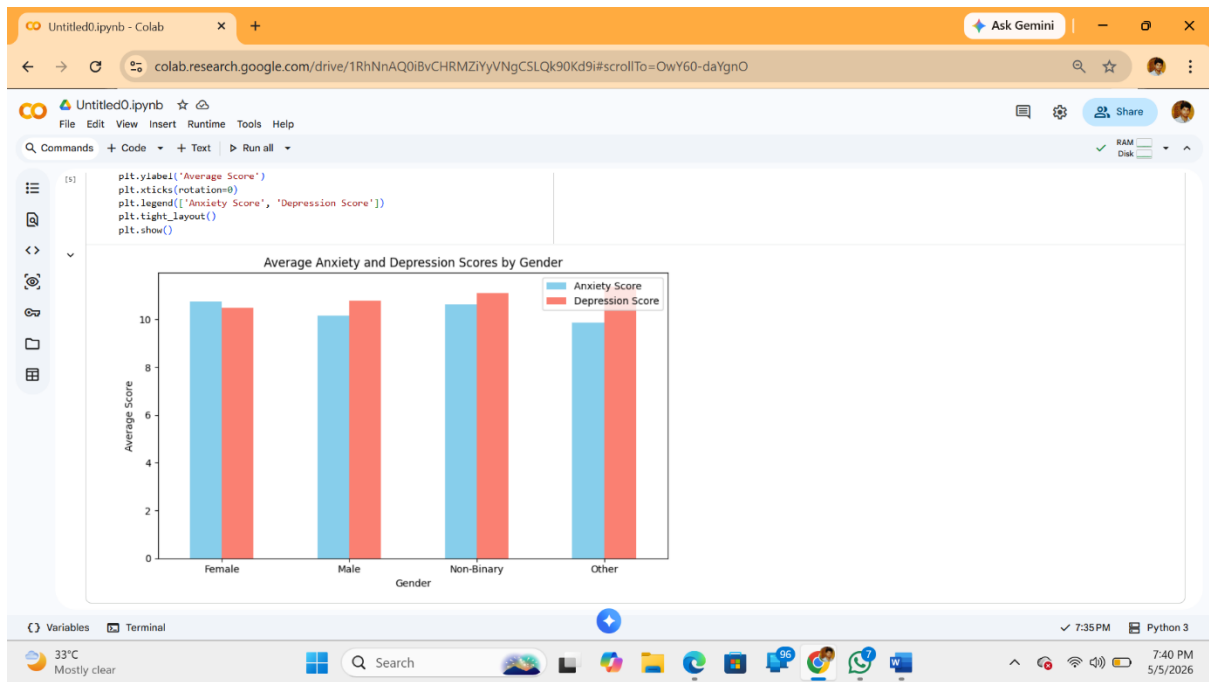


Analysis of Disorder Prevalence by Gender

The bar chart below illustrates the mean anxiety and depression scores categorized by gender.

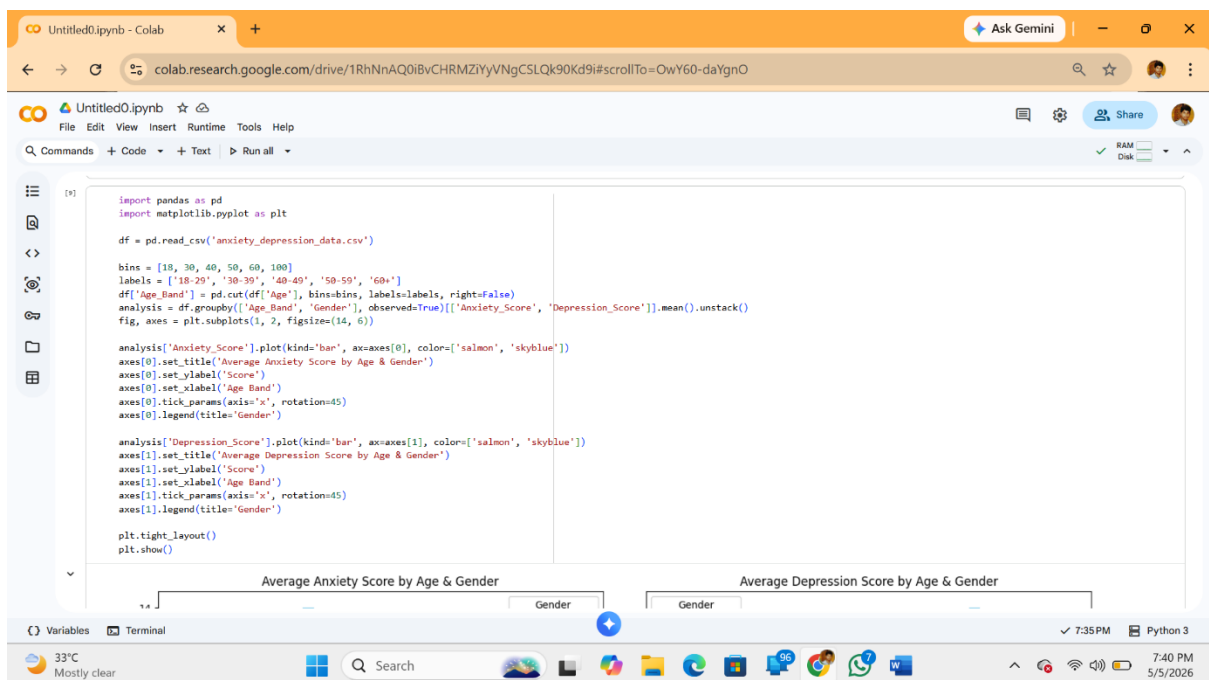
Based on the dataset, we observe the mean scores for each gender group. While scores are generally distributed, this visual breakdown helps highlight whether certain gender cohorts in the survey report higher average levels of anxiety or depression, serving as a primary indicator for further psychological research.

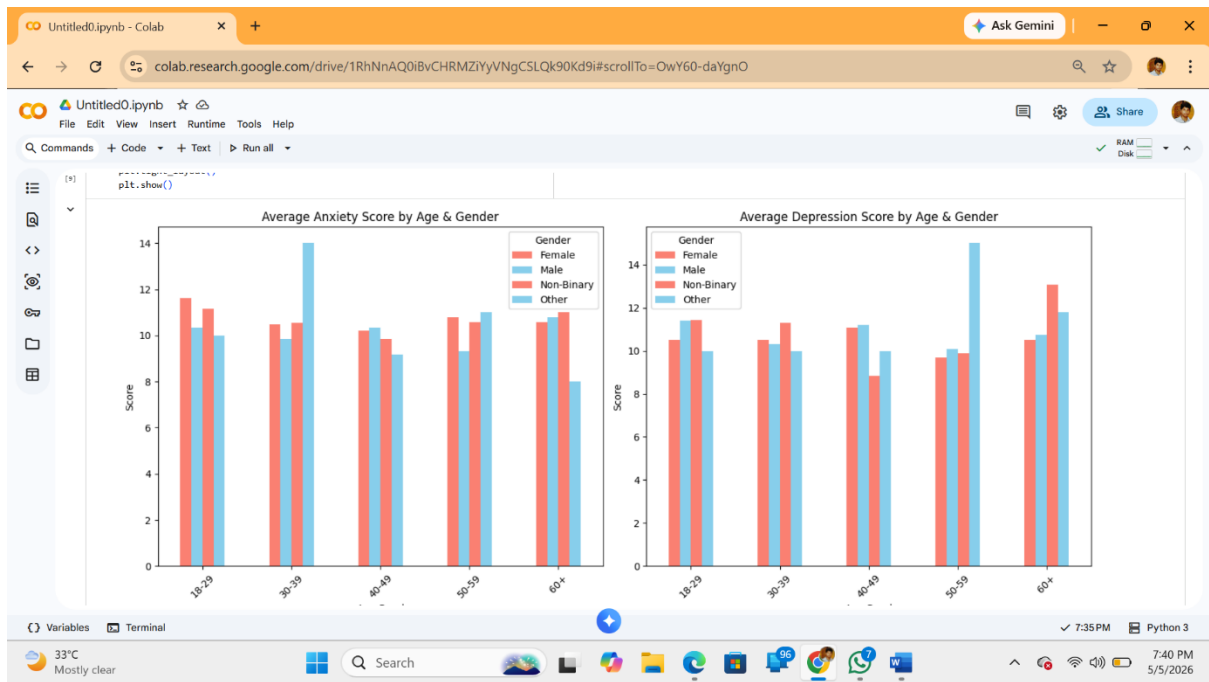




Analysis of Disorder Prevalence by Age Band vs gender

This analysis provides a granular breakdown of disorder prevalence by combining two key demographic dimensions: **Age Band** and **Gender**. By visualizing how Anxiety_Score and Depression_Score vary across different age cohorts and gender groups, we can identify specific demographic intersections that may be at higher risk, moving beyond a single-variable analysis.





Ethnicity:

Upon reviewing the available dataset, **no "Ethnicity" column exists**. Consequently, a breakdown by ethnicity cannot be performed. The analysis below focuses on **Age Band** and **Gender**, which are the available demographic identifiers.

Conclusion:

This analysis shows that mental health severity varies slightly across gender groups. Women tend to have slightly higher percentages in moderate and severe categories, while men show higher values in lower severity levels. The "All Adults" group provides an overall view of the population. Overall, the dataset shows that most individuals fall under lower severity categories, and severe conditions are relatively less common.